

Differential Geometry Exercice 4

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Question 1

Let $M \subset \mathbb{R}^3$ be a two-dimensional, embedded submanifold with the induced Riemannian metric. Let $U \subset \mathbb{R}^3$ be a neighborhood of 0 and let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function with $f(0) = 0$ and $\nabla f(0) = 0$. Let us assume that

$$M \cap U = \{(x, y, z) \in U : z = f(x, y)\} \quad (1)$$

Local orthonormal coframe

The induced metric comes from $\mathbb{R}^3$ Euclidian metric:

$$g_{\mathbb{R}^3} = dx^2 + dy^2 + dz^2 \quad (2)$$

Inside $M \cap U$, the induced metric takes the form:

$$g_{M \cap U} = dx^2 + dy^2 + (f_x dx + f_y dy)^2 = (1 + f_x^2) dx^2 + 2f_x f_y dx dy + (1 + f_y^2) dy^2 \quad (3)$$

Where

$$f_x = \frac{\partial f}{\partial x} \quad f_y = \frac{\partial f}{\partial y} \quad (4)$$

To find a coframe, let us use Cholesky decomposition. Define the two one-forms:

$$\omega_1 = a_x dx \quad \omega_2 = b_x dx + b_y dy \quad (5)$$

Such that:

$$g = \omega_1^2 + \omega_2^2 \quad (6)$$

Opening this up gives:

$$g = (a_x^2 + b_x^2) dx^2 + 2b_x b_y dx dy + b_y^2 dy^2 \quad (7)$$

comparing this to the original metric gives:

$$a_x = \sqrt{1 + \frac{f_x^2}{1 + f_y^2}} \quad b_x = \frac{f_x f_y}{\sqrt{1 + f_y^2}} \quad b_y = \sqrt{1 + f_y^2} \quad (8)$$

And so, and orthonotmal frame is:

$$\omega_1 = \sqrt{1 + \frac{f_x^2}{1 + f_y^2}} dx \quad \omega_2 = \frac{f_x f_y}{\sqrt{1 + f_y^2}} dx + \sqrt{1 + f_y^2} dy \quad (9)$$

One form derivation

Let

$$\omega = \omega_1 + i\omega_2 \quad (10)$$

And let us look for the 1-form φ such that $d\omega = i\varphi \wedge \omega$. Plugging the definition of ω into it gives:

$$d\omega_1 + id\omega_2 = i\varphi \wedge (\omega_1 + i\omega_2) = -\varphi \wedge \omega_2 + i\varphi \wedge \omega_1 \quad (11)$$

Which requires:

$$\begin{cases} d\omega_1 = -\varphi \wedge \omega_2 \\ d\omega_2 = \varphi \wedge \omega_1 \end{cases} \quad (12)$$

Since ω_1, ω_2 is a coframe, φ can be written as its linear combination $\varphi = \varphi_1\omega_1 + \varphi_2\omega_2$:

$$\begin{cases} d\omega_1 = -\varphi_1\omega_1 \wedge \omega_2 \\ d\omega_2 = -\varphi_2\omega_1 \wedge \omega_2 \end{cases} \quad (13)$$

Let us now derive the values of these expressions in terms of dx and dy :

$$\omega_1 \wedge \omega_2 = \sqrt{\left(1 + \frac{f_x^2}{1 + f_y^2}\right) (1 + f_y^2)} dx \wedge dy = \sqrt{1 + f_x^2 + f_y^2} dx \wedge dy \quad (14)$$

$$\begin{aligned} d\omega_1 &= -\frac{\partial a_x}{\partial y} dx \wedge dy = -a_x^{-1} \left(\frac{(1 + f_y^2)f_x f_{xy} - f_x^2 f_y f_{yy}}{(1 + f_y^2)^2} \right) dx \wedge dy = \\ &= -\frac{f_x f_{xy} + f_y^2 f_x f_{xy} - f_x^2 f_y f_{yy}}{(1 + f_y^2)^{3/2} (1 + f_x^2 + f_y^2)^{1/2}} dx \wedge dy = \end{aligned} \quad (15)$$

$$\begin{aligned} &= -\frac{f_x(f_{xy} + f_y^2 f_{xy} - f_x f_y f_{yy})}{(1 + f_y^2)^{3/2} (1 + f_x^2 + f_y^2)^{1/2}} dx \wedge dy \\ d\omega_2 &= \left(\frac{\partial b_y}{\partial x} - \frac{\partial b_x}{\partial y} \right) dx \wedge dy = \\ &= \left(\frac{f_y f_{xy}}{\sqrt{1 + f_y^2}} - \frac{(f_{xy} f_y + f_x f_{yy})(1 + f_y^2) - f_x f_y^2 f_{xy}}{(1 + f_y^2)^{3/2}} \right) dx \wedge dy = \\ &= \left(\frac{(f_{xy} f_y - f_x f_{yy} - f_y f_{xy})(1 + f_y^2) + f_x f_y^2 f_{xy}}{(1 + f_y^2)^{3/2}} \right) dx \wedge dy = \\ &= -\frac{f_x(f_{yy} + f_y^2 f_{yy} + f_y^2 f_{xy})}{(1 + f_y^2)^{3/2}} dx \wedge dy \end{aligned} \quad (16)$$

This gives:

$$\varphi_1 = \frac{f_x(f_{xy} + f_y^2 f_{xy} - f_x f_y f_{yy})}{(1 + f_y^2)^{3/2} (1 + f_x^2 + f_y^2)} \quad \varphi_2 = \frac{f_x(f_{yy} + f_y^2 f_{yy} + f_y^2 f_{xy})}{(1 + f_y^2)^{3/2} (1 + f_x^2 + f_y^2)^{1/2}} \quad (17)$$

Theorema Egregium

Let us now prove Theorema Egregium at the origin, i.e. show that:

$$d\varphi = K\omega_1 \wedge \omega_2 \quad s.t. \quad K = \det \nabla^2 f(0) \quad (18)$$

To show that, let us expand f_x and f_y around the origin (all second derivatives from this point on are calculated at the origin, i.e. $f_{xx} = f_{xx}(0)$ and so forth):

$$f_x \approx f_{xx}x + f_{xy}y \quad f_y \approx f_{xy}x + f_{yy}y \quad (19)$$

Pluggin that into the formulas for φ_1 and φ_2 and keeping only linear terms:

$$\varphi_1 = \frac{f_x(f_{xy} + f_y^2 f_{xy} - f_x f_y f_{yy})}{(1 + f_y^2)^{3/2} (1 + f_x^2 + f_y^2)} \approx f_{xy}(f_{xx}x + f_{xy}y) \quad (20)$$

$$\varphi_2 = \frac{f_x(f_{yy} + f_y^2 f_{yy} + f_y^2 f_{xy})}{(1 + f_y^2)^{3/2} (1 + f_x^2 + f_y^2)^{1/2}} \approx f_{yy}(f_{xx}x + f_{xy}y) \quad (21)$$

Which gives the form:

$$\begin{aligned} \varphi &= f_{xy}(f_{xx}x + f_{xy}y)\omega_1 + f_{yy}(f_{xx}x + f_{xy}y)\omega_2 = \\ &= (f_{xx}x + f_{xy}y) \left(f_{yy} \frac{f_x f_y}{\sqrt{1 + f_y^2}} + f_{xy} \sqrt{1 + \frac{f_x^2}{1 + f_y^2}} \right) dx + f_{yy}(f_{xx}x + f_{xy}y) \sqrt{1 + f_y^2} dy \approx \\ &\approx f_{xy}(f_{xx}x + f_{xy}y)dx + f_{yy}(f_{xx}x + f_{xy}y)dy \end{aligned} \quad (22)$$

Deriving this, we get:

$$\begin{aligned} d\varphi &= \left(\frac{\partial}{\partial x} (f_{yy}(f_{xx}x + f_{xy}y)) - \frac{\partial}{\partial y} (f_{xy}(f_{xx}x + f_{xy}y)) \right) dx \wedge dy = \\ &= (f_{xx}f_{yy} - f_{xy}^2) dx \wedge dy = \det \nabla^2 f(0) dx \wedge dy \end{aligned} \quad (23)$$

But, in the origin:

$$\omega_1 \wedge \omega_2 = \sqrt{1 + f_x^2 + f_y^2} dx \wedge dy = dx \wedge dy \quad (24)$$

Thus:

$$d\varphi = \det \nabla^2 f(0) \omega_1 \wedge \omega_2 \quad (25)$$